Quick § 8.1, 8.2, 8.3:

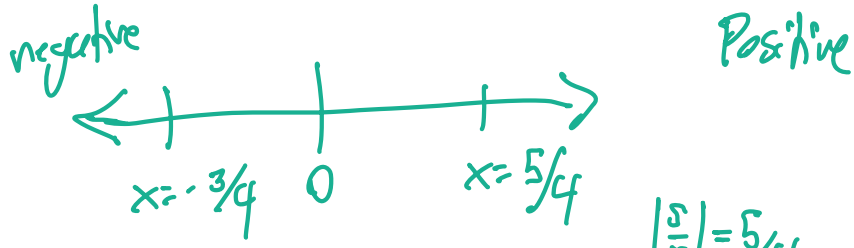
Let F $(= \mathbb{Q}, \mathbb{R}, \mathbb{C})$ be an ordered field and suppose we have $<, >, \leq,$ and $\geq$ behaving in ways we expect.

Theorem 6 (Trichotomy). If $x, y \in F$, then exactly one of the following occurs:

i) $x = y$, ii) $y < x$, or iii) $x < y$.

Jumping to § 8.2. The <u>absolute value</u> $|x|$ of $x \in F$ is defined to be $|x| \overset{\text{def}}{=} \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.

"The distance from x to 0."

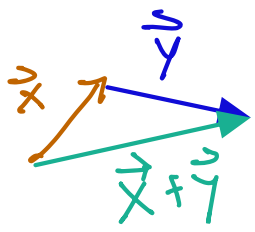


$|\frac{5}{4}| = 5/4$

$|x| = |-3/4| = -(-3/4) = 3/4$

"$|x-y|$ the distance between x and y."

<u>Theorem 30.</u> (The Triangle Inequality) If $x, y \in F$, then $|x+y| \leq |x| + |y|$.

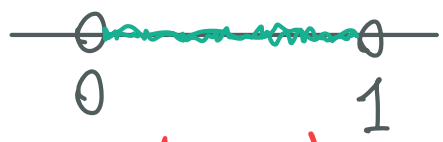


$\|\vec{x}\| + \|\vec{y}\| \geq \|\vec{x}+\vec{y}\|$

<u>Proof</u>. If $x \geq 0$, then $|x| = x$. If $x < 0$, then $|x| > x$. Thus $|x| \geq x$. Hence $x+y \leq |x|+y \leq |x|+|y|$. Likewise $-x-y \leq |x|-y \leq |x|+|y|$. If $x+y \geq 0$, then $|x+y| = x+y \leq |x|+|y|$. If $x+y<0$, then $|x+y| = -x-y \leq |x|+|y|$. Therefore, $|x+y| \leq |x|+|y|$. ✓

Def. 7 (Intervals) For $x, y \in F$ with $x<y$ define $(x,y) \overset{\text{def}}{=} \{z \in F : x < z < y\}$.

For example, $(0,1)$ in $\mathbb{R}$ is



Theorem 32 (Density) "Between any two rational numbers is another rational number" Let $x, y \in F$ with $x<y$. Then $\exists$ $z \in F$ with $x<z<y$. In other words, $(x,y) \neq \emptyset$.

$$(x,y) = \{z \in F : x<z<y\}$$

$\frac{x+y}{2} = z$.

$$\left(\frac{314159}{100000}, \frac{22}{7}\right)$$

In $\mathbb{Z}$,

$$(1,2) = \{z \in \mathbb{Z} : 1<z<2\} = \emptyset$$

Proof. Consider $z = \frac{x+y}{2} = 2^{-1}(x+y)$. $\frac{1+2}{2} = \frac{3}{2}$

Since $x<y$, $x+x<x+y<y+y$. Thus $2x<x+y<2y$.

As $2 \neq 0$, $2^{-1} = \frac{1}{2} \in F$. Since $2>0$, $\frac{1}{2}>0$. Thus

$\frac{1}{2}\cdot 2x < \frac{1}{2}(x+y) < \frac{1}{2}\cdot 2y$. Hence $x < \frac{x+y}{2} < y$

§8.3 The Archimedean Property.

Def. 10. F is <u>Archimedean</u> if $\mathbb{Q} \subset F$ as an ordered subfield and if $x, y \in F$ with $x>0$, then $\exists\, n \in \mathbb{N} \subset F$ such that $nx \geq y$.

Choose $x=1$ to see that the natural numbers are unbounded in F.

Class. really big # 20 $\quad 1/401$ $\quad$ take any $n \in \mathbb{N}$ with $n \geq 20 \cdot 401$

Then $n \cdot \frac{1}{401} \geq 20$

Theorem 37. If F is an Archimedean ordered field then $\mathbb{Q}$ is dense in F. In other words,

if $x<y$ in F, then there exists $r \in \mathbb{Q}$ such that $x<r<y$.

Ch. 8 §5: Infinite Sequences and limits

Def. An <u>infinite sequence with values in F</u> is a function whose domain is the natural numbers and whose codomain is F. Write a_i to denote the image of $i \in \mathbb{N}$ under this function.

we often augment/restrict this

Examples. $a_i = i$ is $\{0, 1, 2, 3, \dots\}$ in $\mathbb{Q}$

$a_i = \frac{1}{i}$ with $i>0$ is $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$

$a_i = \frac{1}{i+1}$

$b_i = \frac{-1}{i^2}$ with $i>0$ is $\{-1, -1/4, -1/9, -1/16, \dots\}$

$c_i = (-1)^i \frac{1}{i^2-i}$ for $i \geq 1$ is $\{(-1)^2 \frac{1}{2^2-2} = \frac{1}{2}, -1/6, \frac{1}{12}, \frac{-1}{20}, \dots\}$.

Problem 1: $\sqrt{2}$ and other friends are not in $\mathbb{Q}$

Solution: Take sequences of rationals

$$1, \frac{14}{10}, \frac{141}{100}, \frac{1414}{1{,}000}, \frac{14142}{10{,}000}, \ldots \qquad \sqrt{2}$$

Problem 2: There are multiple sequences that approach the same thing

$$1, \frac{14}{10}, \frac{14}{10}, \frac{141}{100}, \frac{141}{100}, \frac{1414}{1{,}000}, \frac{1414}{1000}, \frac{1414}{1000}, \frac{14142}{10{,}000}, \ldots \quad \sqrt{2}$$

$$1, 1+\frac{1}{3}, 1+\frac{1}{3}+\frac{2}{27}, 1+\frac{1}{3}+\frac{2}{27}+\frac{1}{243}, \ldots \qquad \sqrt{2}$$

Solution: Equivalence classes of sequences

$$(a_i) \sim (b_i) \overset{(\text{def})}{\text{ if }} \lim_{i \to \infty} a_i - b_i = 0$$

Problem 3: There are sequences that shouldn't converge

$$1, 2, 3, 4, 5, \ldots \qquad 1, -1, 1, -1, 1, -1, \ldots$$

Solution: Take only equivalence classes of sequences that "should" converge.

Define a sequence (a_i) to be <u>Cauchy</u> if $\forall\, \varepsilon > 0\ \exists\, N \in \mathbb{N}$ such that if $i, j \geq N$,

$$|a_i - a_j| < \varepsilon.$$

Math Debt.

Prove sequences that converge are Cauchy.

Ch. 10, Defs 1 & 2 (The Real Numbers)

If (a_i) is a Cauchy sequence in $\mathbb{Q}$, then define $[(a_i)]$ to be the equivalence class of (a_i). I.e., $[(a_i)] = \{(b_i) : (b_i) \sim (a_i)\}$.

Define the <u>real numbers</u>, denoted by $\mathbb{R}$, to be

$$\mathbb{R} \overset{\text{def}}{=} \{[(a_i)] : (a_i) \text{ is a Cauchy sequence in } \mathbb{Q}\}.$$

Def. 3 ($+$ & $\cdot$ on $\mathbb{R}$)

$$[(a_i)_{i \geq n_0}] + [(b_i)_{i \geq m_0}] = [(a_i + b_i)_{i \geq \max(n_0, m_0)}]$$

$$[(a_i)_{i\geq n_0}]\cdot[(b_i)_{i\geq m_0}] = [(a_i\cdot b_i)_{i\geq \max(n_0,m_0)}].$$

Do Operations with sequences work well?

All from Chapter 8: <u>Thm 4.4</u>. Suppose $a_i = a$ for all i. Then $\lim_{i\to\infty} a_i = a$.

$\frac{1}{2} = [(\frac{1}{2})]$

<u>Thm 4.5</u>. If $a_i \leq b_i$ for all i sufficiently large, then $\lim_{i\to\infty} a_i \leq \lim_{i\to\infty} b_i$.

Calc limit laws forever!

Define a sequence (a_i) to be <u>Cauchy</u> if $\forall\, \varepsilon>0\ \exists\, N\in\mathbb{N}$ such that if $i,j\geq N$, $|a_i - a_j| < \varepsilon$.

sequences that should converge.

Ch. 9, Thm. 15. Suppose $\lim_{i\to\infty} a_i = L$. Then (a_i) is Cauchy.

Proof. Let $\varepsilon>0$. Since $\lim_{i\to\infty} a_i = L$ $\exists\, N\in\mathbb{N}$ s.t. if $i\geq N$, then $|a_i - L| < \frac{\varepsilon}{2}$.

Now if $i,j\geq N$, then

$$|a_i - a_j| = |a_i - L + L - a_j| \le |a_i - L| + |L - a_j| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

$+L-L$ trick

Δ-inequality

Thus (a_i) is Cauchy. $\square$

Ch. 9

Thm. 16 If $(a_i) \sim (b_i)$ and (a_i) is Cauchy, then (b_i) is Cauchy.

Scratch. want $|b_i - b_j| < \varepsilon$

Have $|a_i - a_j|$ small and $|a_i - b_i|$ small

$$|b_i - b_j| = |b_i - a_i + a_i - a_j + a_j - b_j| \le \underbrace{|b_i - a_i|}_{\varepsilon/3} + \underbrace{|a_i - a_j|}_{\varepsilon/3} + \underbrace{|a_j - b_j|}_{\varepsilon/3}$$

Proof. Let $\varepsilon > 0$. Since $(a_i) \sim (b_i)$ & (a_i) is Cauchy

$\exists\ N_1 \in \mathbb{N}$ s.t. if $i \ge N_1$, then $|a_i - b_i| < \frac{\varepsilon}{3}$ and

$\exists\ N_2 \in \mathbb{N}$ s.t. if $i, j \ge N_2$, then $|a_i - a_j| < \frac{\varepsilon}{3}$.

Let $N = \max(N_1, N_2)$. If $i, j \ge N$, then

$$|b_i - b_j| = |b_i - a_i + a_i - a_j + a_j - b_j| \le |b_i - a_i| + |a_i - a_j| + |a_j - b_j|$$

Δ-inequality

$$< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus (b_i) is Cauchy! $\square$